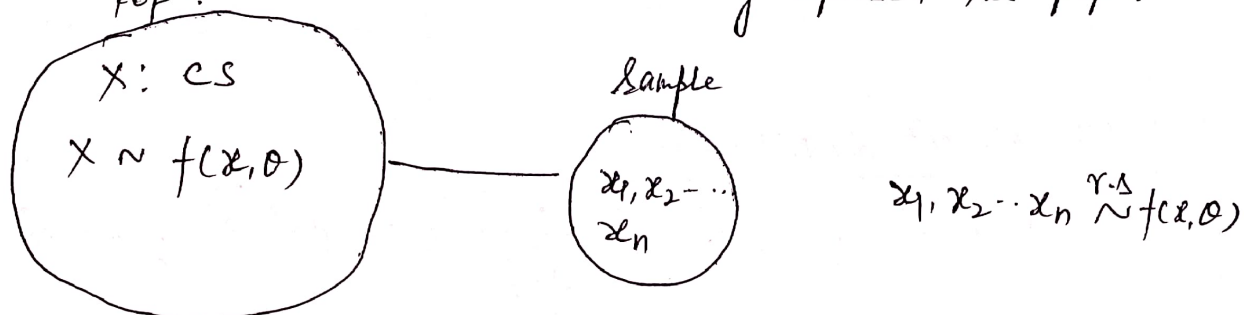


Sampling Distributions

Page 1

Population: A group of individuals defined according to the aim of the study. Ex. Human popⁿ, popⁿ of households, popⁿ of electric bulbs produced by an industry, popⁿ of vehicles etc.

Sample: A subset of popⁿ which truly represent the popⁿ.



Sample values such as sample mean $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$,
sample mean square/sample variance $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
order statistics $x_{(1)}, x_{(2)}, \dots, x_{(n)}$

are the random variables and their prob distributions are known as sampling distributions.

Specific Sampling Distributions:

Chi-Square variate (χ^2 -variate): The square of a standard normal variate is known as a χ^2 -variate with 1 d.f. (degree of freedom).

If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$

and $Z^2 = \left(\frac{X - \mu}{\sigma}\right)^2$ is a χ^2 variate with 1 d.f. and denoted as χ_1^2 .

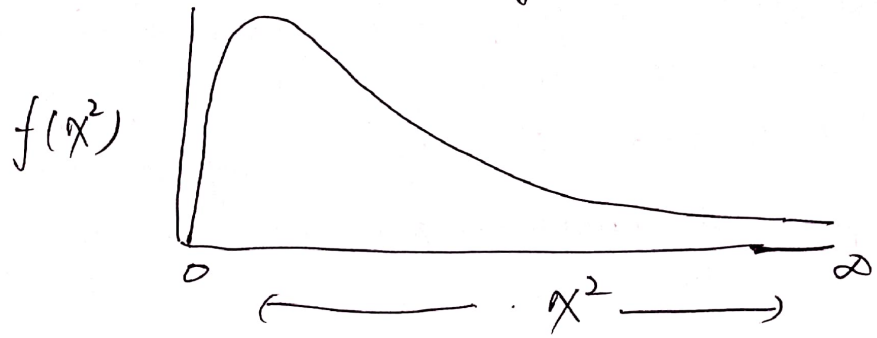
In general, if X_i ($i = 1, 2, \dots, n$) are n independent normal variates with mean μ_i and variance σ_i^2 , ($i = 1, 2, \dots, n$), then

$$\chi^2 = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i}\right)^2 \sim \chi_n^2.$$

The prob density function of χ^2 -variate with n d.f. is given as

$$f(\chi^2) = \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} e^{-\frac{\chi^2}{2}} (\chi^2)^{\frac{n}{2}-1}; \quad 0 \leq \chi^2 < \infty.$$

The prob curve of χ^2 variate is +vely skewed and given as



MGF of χ^2 distribution

$$\chi^2 \sim \chi_n^2 ; \quad f(x^2) = \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} e^{-\frac{x^2}{2}} (x^2)^{\frac{n}{2}-1} ; \quad 0 \leq x^2 < \infty$$

$$M_{\chi^2}(t) = E e^{t\chi^2} = \int_0^{\infty} e^{t\chi^2} f(x^2) dx^2 = \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \int_0^{\infty} e^{t\chi^2} e^{-\frac{x^2}{2}} (x^2)^{\frac{n}{2}-1} dx^2$$

$$= \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \int_0^{\infty} e^{-(\frac{1-2t}{2})x^2} (x^2)^{\frac{n}{2}-1} dx^2$$

$$= \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \frac{\Gamma(\frac{n}{2})}{\left[\left(\frac{1-2t}{2}\right)\right]^{\frac{n}{2}}}$$

$$= (1-2t)^{-\frac{n}{2}} ; \quad |2t| < 1$$

(using gamma integral)

$$\int_0^{\infty} e^{-ax} x^{n-1} dx = \frac{\Gamma(n)}{a^n}$$

$$h_1' = \text{Mean} = \frac{dM_{\chi^2}(t)}{dt} \Big|_{t=0} ; \quad \frac{dM_{\chi^2}(t)}{dt} = -\frac{n}{2} (1-2t)^{-\frac{n}{2}-1} (-2)$$

Putting $t=0$, Mean = n

$$h_2' = \frac{d^2 M_{\chi^2}(t)}{dt^2} \Big|_{t=0} ; \quad \frac{d^2 M_{\chi^2}(t)}{dt^2} = \left(-\frac{n}{2}\right) \left(-\frac{n}{2}-1\right) (1-2t)^{-\frac{n}{2}-2} (-2)(-2)$$

Putting $t=0$, $h_2' = n(n+2)$

$$\therefore \text{variance } (h_2) = h_2' - h_1'^2 = n(n+2) - n^2 = n^2 + 2n - n^2 = 2n$$

Hence $\boxed{\text{Mean} = n} + \boxed{\text{Variance} = 2n}$

χ^2 holds additive property

$\chi_1^2 \sim \chi_{n_1}^2$, $\chi_2^2 \sim \chi_{n_2}^2$... $\chi_k^2 \sim \chi_{n_k}^2$ are independent χ^2 variates, then $\sum_{i=1}^k \chi_i^2$ will follow $\chi_{n_1+n_2+\dots+n_k}^2$.

Proof $\therefore M_{\chi_i^2}(t) = (1-2t)^{-\frac{n_i}{2}}$ ($i=1, 2, \dots, k$)

$$M_{\sum_{i=1}^k \chi_i^2}(t) = \prod_{i=1}^k M_{\chi_i^2}(t) = (1-2t)^{-\frac{1}{2} \sum_{i=1}^k n_i}$$

which is the m.g.f. of χ^2 variate with $\sum_{i=1}^k n_i$ d.f.

Hence $\sum_{i=1}^k \chi_i^2 \sim \chi_{n_1+n_2+\dots+n_k}^2$.

Ex If X_1 and X_2 are two independent χ^2 -variates with n_1 and n_2 d.f. respectively, then $\frac{X_1}{X_2}$ is a $\beta_2\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$ variate.

Solⁿ: $X_1 \sim \chi_{n_1}^2$ and $X_2 \sim \chi_{n_2}^2$ are independent χ^2 variates with n_1 and n_2 d.f.; the joint p.d.f. of X_1 and X_2 is given as

$$\begin{aligned} f(X_1, X_2) &= f_1(X_1) \cdot f_2(X_2) \\ &= \frac{1}{2^{\frac{n_1}{2}} \sqrt{\frac{n_1}{2}}} e^{-\frac{x_1}{2}} \frac{x_1^{\frac{n_1}{2}-1}}{2^{\frac{n_2}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{x_2}{2}} \frac{x_2^{\frac{n_2}{2}-1}}{2^{\frac{n_2}{2}} \sqrt{\frac{n_2}{2}}} \\ &= \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{x_1+x_2}{2}} \frac{x_1^{\frac{n_1}{2}-1}}{2^{\frac{n_1}{2}} \sqrt{\frac{n_1}{2}}} \frac{x_2^{\frac{n_2}{2}-1}}{2^{\frac{n_2}{2}} \sqrt{\frac{n_2}{2}}} \quad 0 \leq (x_1, x_2) < \infty \end{aligned}$$

Let us make a transformation

$$U = \frac{x_1}{x_2} \quad \text{and} \quad V = x_2 \quad \text{so that} \quad x_1 = UV \quad \text{and} \quad x_2 = V$$

Jacobian of transformation J is given by $J = \frac{\partial(x_1, x_2)}{\partial(u, v)}$

Thus the joint distribution of random variables U and V becomes:

$$\begin{aligned} g(u, v) &= f(x_1, x_2) \cdot |J| \\ &= \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{(1+u)v}{2}} (uv)^{\frac{n_1}{2}-1} v^{\frac{n_2}{2}-1} \cdot v \\ &= \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{(1+u)v}{2}} (uv)^{\frac{n_1}{2}-1} v^{\frac{n_2}{2}} \end{aligned}$$

$$\begin{aligned} J &= \begin{vmatrix} \frac{\partial x_1}{\partial u} & \frac{\partial x_1}{\partial v} \\ \frac{\partial x_2}{\partial u} & \frac{\partial x_2}{\partial v} \end{vmatrix} \\ &= \begin{vmatrix} v & u \\ 0 & 1 \end{vmatrix} = v \\ |J| &= v. \end{aligned}$$

$$\Rightarrow g(u, v) = \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{(1+u)v}{2}} u^{\frac{n_1}{2}-1} v^{\frac{n_1+n_2}{2}-1}$$

$0 < u < \infty$
 $0 < v < \infty$

$$\begin{aligned} \therefore g_1(u) &= \int_0^{\infty} g(u, v) dv \\ &= \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} u^{\frac{n_1}{2}-1} \int_0^{\infty} e^{-\frac{(1+u)v}{2}} v^{\frac{n_1+n_2}{2}-1} dv \\ &= \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} \cdot u^{\frac{n_1}{2}-1} \cdot \frac{\sqrt{\frac{n_1+n_2}{2}}}{\left(\frac{1+u}{2}\right)^{\frac{n_1+n_2}{2}}} \quad (\text{using gamma integration}) \\ &= \frac{\sqrt{\frac{n_1+n_2}{2}}}{\sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} \cdot \frac{u^{\frac{n_1}{2}-1}}{(1+u)^{\frac{n_1+n_2}{2}}} ; \quad 0 < u < \infty \\ &= \frac{1}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{u^{\frac{n_1}{2}-1}}{(1+u)^{\frac{n_1+n_2}{2}}} ; \quad 0 < u < \infty \end{aligned}$$

which is the p.d.f. of $\beta_2\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$

Hence $U = \frac{X_1}{X_2} \sim \beta_2\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$

Ex. If X_1 and X_2 are independent χ^2 variates with n_1 and n_2 d.f. respectively, then $U = \frac{X_1}{X_1+X_2}$ and $V = X_1+X_2$ are independently distributed, U as a $\beta_1\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$ variate and V as a χ^2 variate with n_1+n_2 d.f.

Solⁿ: $X_1 \sim \chi_{n_1}^2$ and $X_2 \sim \chi_{n_2}^2$ are indep χ^2 variates

$$\therefore f(x_1, x_2) = \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{(x_1+x_2)}{2}} \frac{n_1-1}{2} \frac{n_2-1}{2}, \quad 0 \leq (x_1, x_2) < \infty$$

Let us transform to u and v defined as follows:

$$u = \frac{x_1}{x_1+x_2} \quad \text{and} \quad v = x_1+x_2 \quad \text{so that} \quad x_1 = uv \quad \text{and}$$

$$\text{as} \quad 0 < u < 1, \quad 0 < v < \infty \quad x_2 = v - x_1 = (1-u)v$$

Jacobian of transformation J is

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial u} & \frac{\partial x_1}{\partial v} \\ \frac{\partial x_2}{\partial u} & \frac{\partial x_2}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} v & u \\ -v & 1-u \end{vmatrix}$$

$$= v(1-u) + uv = v$$

$$|J| = v$$

Hence

$$g(u, v) = |(x_1, x_2) J|$$

$$= \frac{1}{\frac{n_1+n_2}{2} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{v}{2}} (uv)^{\frac{n_1}{2}-1} [(1-u)v]^{\frac{n_2}{2}-1} \cdot v$$

$$= \frac{1}{\frac{n_1+n_2}{2} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} u^{\frac{n_1}{2}-1} (1-u)^{\frac{n_2}{2}-1} e^{-\frac{v}{2}} v^{\frac{n_1+n_2}{2}-1}$$

$$= \left[\frac{\sqrt{\frac{n_1+n_2}{2}}}{\sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} u^{\frac{n_1}{2}-1} (1-u)^{\frac{n_2}{2}-1} \right] \left[\frac{1}{\frac{n_1+n_2}{2} \sqrt{\frac{n_1+n_2}{2}}} e^{-\frac{v}{2}} v^{\frac{n_1+n_2}{2}-1} \right]$$

$$= g_1(u) \cdot g_2(v)$$

where $g_1(u) = \frac{1}{\beta(\frac{n_1}{2}, \frac{n_2}{2})} u^{\frac{n_1}{2}-1} (1-u)^{\frac{n_2}{2}-1}$ is the p.d.f. of $\beta_1(\frac{n_1}{2}, \frac{n_2}{2})$

and $g_2(v) = \frac{1}{\frac{n_1+n_2}{2} \sqrt{\frac{n_1+n_2}{2}}} e^{-\frac{v}{2}} (v)^{\frac{n_1+n_2}{2}-1}$ is the p.d.f. of $\chi^2_{n_1+n_2}$.

Hence $U = \frac{X_1}{X_1 + X_2} \sim \beta_1\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$

and $V = X_1 + X_2 \sim \chi^2_{n_1+n_2}$.

Remarks: (i) If $X \sim \chi^2_{n_1}$ and $Y \sim \chi^2_{n_2}$ are independent χ^2 -variables.

(i) $X + Y \sim \chi^2_{n_1+n_2}$

(ii) $\frac{X}{Y} \sim \beta_2\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$

(iii) $\frac{X}{X+Y} \sim \beta_1\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$

2) If $X_1, X_2, \dots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$

$$(i) \quad \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \& \quad Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

$$(ii) \quad \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 \sim \chi_{n-1}^2$$

(iii) $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\frac{(n-1)S^2}{\sigma^2}$ are independently distributed.
 where $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

t-distribution

If $X_1, X_2, \dots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$

then the Student's 't' is defined as

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$f(t) = \frac{1}{\sqrt{v} B\left(\frac{1}{2}, \frac{v}{2}\right) \left[1 + \frac{t^2}{v}\right]^{\frac{v+1}{2}}}; -\infty < t < \infty$$

where $v = n-1$

Remarks: ① If a statistic t follows Student's t -distribution with n d.f. will be abbreviated as $t \sim t_n$.

(ii) If we take $n=1$, we get

$$f(t) = \frac{1}{B\left(\frac{1}{2}, \frac{1}{2}\right) (1+t^2)} = \frac{1}{\pi} \cdot \frac{1}{(1+t^2)}, -\infty < t < \infty$$

which is the p.d.f. of standard Cauchy distⁿ. $\frac{1}{2} = \sqrt{\pi}$

Fisher's 't' : It is the ratio of a standard normal variate to the square root of an independent χ^2 -variate divided by its degrees of freedom.

i.e. If $Z \sim N(0,1)$ and $\chi^2 \sim \chi_n^2$

then Fisher's t is given as

$$t = \frac{Z}{\sqrt{\chi^2/n}} \sim t_n.$$

$$f(t) = \frac{1}{\sqrt{n} \beta\left(\frac{1}{2}, \frac{n}{2}\right) \left[1 + \frac{t^2}{n}\right]^{\frac{n+1}{2}}}; -\infty < t < \infty$$

Moments of t distribution

Since $f(t)$ is symmetrical about the line $t=0$, all the moments of odd order about origin vanish. i.e.

$$\mu'_{2r+1} = 0; r = 0, 1, 2, \dots$$

In particular $\mu'_1 = 0$

Hence the central order moments coincide with moments about origin. i.e. $\mu_{2r+1} = 0, (r = 0, 1, 2, \dots)$

$$\text{Hence } \mu_{2r} = \mu'_{2r} = \int_{-\infty}^{\infty} t^{2r} f(t) dt = \int_0^{\infty} 2t^{2r} f(t) dt$$

$$= \frac{n^r}{\beta\left(\frac{1}{2}, \frac{n}{2}\right)} \beta\left(\frac{n}{2} - r, r + \frac{1}{2}\right); n > 2r.$$

after simplification we will have

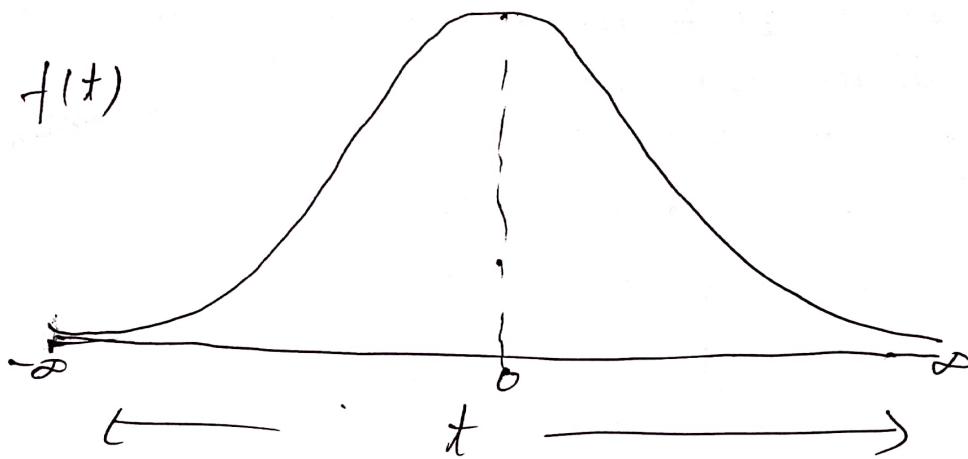
$$\mu_{2r} = n^r \frac{(2r-1)(2r-3) \dots 3 \cdot 1}{(n-2)(n-4) \dots (n-2r)}; \frac{n}{2} > r$$

$$\text{In particular } \mu_2 = \frac{n}{n-2}, n > 2$$

$$\mu_4 = n^2 \cdot \frac{3 \cdot 1}{(n-2)(n-4)} = \frac{3n^2}{(n-2)(n-4)}; n > 4$$

$$\text{Hence } \beta_1 = \frac{\mu_2^2}{\mu_4} = 0, \quad \beta_2 = \frac{\mu_4}{\mu_2^2} = 3 \left(\frac{n-2}{n-4} \right), n > 4$$

$$\text{as } n \rightarrow \infty \quad \beta_2 = 3.$$



F-distribution :

If X and Y are two independent χ^2 variates with ν_1 and ν_2 degrees of freedom respectively then F -variate is defined as

$$F = \frac{X/\nu_1}{Y/\nu_2} \sim F(\nu_1, \nu_2)$$

In other words, F is defined as the ratio of two independent χ^2 variates divided by their respective degrees of freedom and it follows

Snedecor's F -distribution with (ν_1, ν_2) d.f. with prob density function is given by

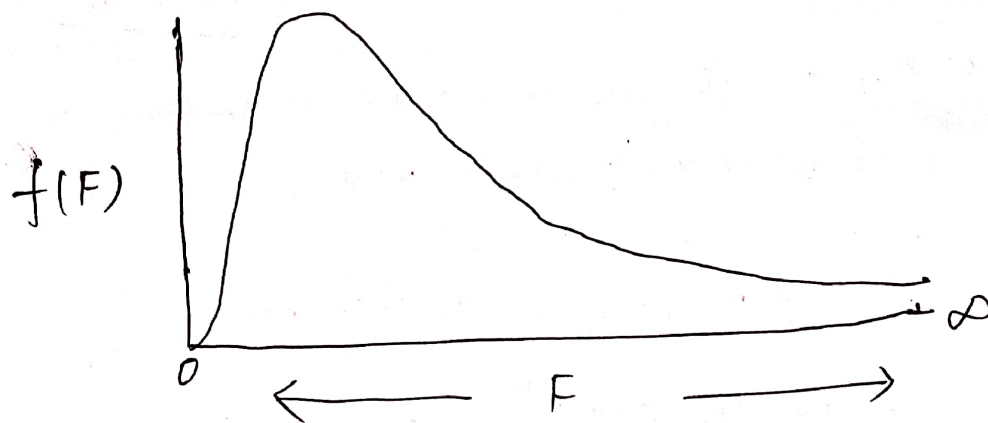
$$f(F) = \frac{\left(\frac{\nu_1}{\nu_2}\right)^{\frac{\nu_1}{2}}}{\beta\left(\frac{\nu_1}{2}, \frac{\nu_2}{2}\right)} \frac{F^{\frac{\nu_1}{2}-1}}{\left[1 + \frac{\nu_1}{\nu_2} F\right]^{\frac{\nu_1+\nu_2}{2}}}, \quad 0 \leq F < \infty$$

Remarks : (i) The sampling distⁿ of F -Statistic does not involve any popⁿ parameters and depends only on the degrees of freedom ν_1 and ν_2 .

(ii) $F \sim F(\nu_1, \nu_2)$

(iii) $F = \frac{X/\nu_1}{Y/\nu_2} \sim F(\nu_1, \nu_2)$ then $\frac{1}{F} = \frac{Y/\nu_2}{X/\nu_1} \sim F(\nu_2, \nu_1)$

(iv) Mean $k_1' = \frac{\nu_2}{\nu_2-2}$, $\nu_2 > 2$, variance $k_2' = \frac{2\nu_2^2(\nu_2+\nu_1-2)}{\nu_1(\nu_2-2)^2(\nu_2-4)}$, $\nu_2 > 4$.



ORDER STATISTICS

Let $X_1, X_2, \dots, X_n$ be n independently and identically distributed (i.i.d) variates, each with cumulative distribution function $F(x)$. If these variables are arranged in ascending order of magnitude and then written as $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$, we call $X_{(r)}$ as the r th order statistic, $r = 1, 2, \dots, n$. The $X_{(r)}$'s because of the inequality relations among them are necessarily dependent.

Remark : If we write these ordered values as

$$Y_1 \leq Y_2 \leq \dots \leq Y_n, \text{ then}$$

$$Y_r = X_{(r)} = r\text{th smallest of } X_1, X_2, \dots, X_n$$

$$Y_1 = X_{(1)} = \text{The smallest of } X_1, X_2, \dots, X_n$$

$$Y_n = X_{(n)} = \text{The largest of } X_1, X_2, \dots, X_n$$

Cumulative Distribution Function of a Single order Statistic

Let $F_r(x)$, $r = 1, 2, \dots, n$ denote the c.d.f. of the r th order statistic $X_{(r)}$. Then the c.d.f. $F_n(x)$ of the largest order statistic $X_{(n)}$ is given by:

$$F_n(x) = P[X_{(n)} \leq x] = P(X_i \leq x; i = 1, 2, \dots, n)$$

$$= P(X_1 \leq x \cap X_2 \leq x \cap \dots \cap X_n \leq x)$$

$$= P(X_1 \leq x) P(X_2 \leq x) \dots P(X_n \leq x) \quad (X_i \text{'s are independent})$$

$$\boxed{F_n(x) = [F(x)]^n} \quad \text{, since } X_1, X_2, \dots, X_n \text{ are i.i.d.}$$

The c.d.f. of $X_{(1)}$ i.e. of smallest (first) order statistic is given by

$$\begin{aligned} F_1(x) &= P(X_{(1)} \leq x) = 1 - P(X_{(1)} > x) \\ &= 1 - P(X_i > x; i=1, 2, \dots, n) \\ &= 1 - \prod_{i=1}^n P(X_i > x) = 1 - \prod_{i=1}^n (1 - P(X_i \leq x)) \\ &= 1 - [1 - F(x)]^n, \text{ Since } X_1, X_2, \dots, X_n \text{ are i.i.d. r.v.s.} \end{aligned}$$

In general, the c.d.f. $F_r(x)$ of the r th order statistic $X_{(r)}$ is given by

$$\begin{aligned} F_r(x) &= P[X_{(r)} \leq x] = P[\text{At least } r \text{ of the } X_i\text{'s are } \leq x] \\ &= \sum_{j=r}^n P[\text{Exactly } j \text{ of the } n, X_i\text{'s are } \leq x] \end{aligned}$$

$$F_r(x) = \sum_{j=r}^n \binom{n}{j} F^j(x) (1-F(x))^{n-j} \quad \text{by using binomial prob. model.}$$

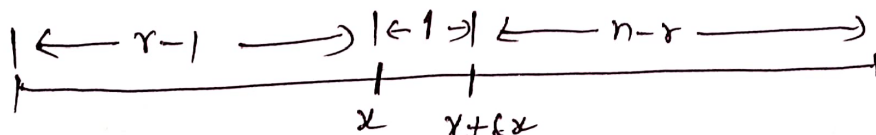
Probability Density Function (p.d.f.) of a single order statistic

Let $X_1, X_2, \dots, X_n$ are i.i.d. r.v.s with p.d.f. $f(x) = F'(x)$.

If $f_r(x)$ denotes the p.d.f. of $X_{(r)}$, then

$$f_r(x) = \lim_{\delta x \rightarrow 0} \frac{P[x < X_{(r)} \leq x + \delta x]}{\delta x} \quad \text{--- (A)}$$

The event $E: x < X_{(r)} \leq x + \delta x$ can materialise as follows:



$X_i \leq x$ for $(r-1)$ of the X_i 's

and $x < X_i \leq x + \delta x$ for one X_i

and $X_i > x + \delta x$ for the remaining $(n-r)$ of the X_i 's

Hence by the multinomial prob. law we have

$$P(x < X_{(r)} \leq x + \delta x) = \frac{n!}{(r-1)! 1! (n-r)!} p_1^{r-1} p_2 p_3^{n-r}$$

where $p_1 = P(X_i \leq x) = F(x)$

$p_2 = P(x < X_i \leq x + \delta x) = F(x + \delta x) - F(x)$

$p_3 = P(X_i > x + \delta x) = 1 - P(X_i \leq x + \delta x) = 1 - F(x + \delta x)$

Hence

$$f_r(x) = \frac{L_n}{L_{r-1} L_{n-r}} F^{r-1}(x) \lim_{\delta x \rightarrow 0} \left[\frac{F(x+\delta x) - F(x)}{\delta x} \right] \cdot \lim_{\delta x \rightarrow 0} [1 - F(x+\delta x)]^{n-r}$$

$$f_r(x) = \frac{L_n}{L_{r-1} L_{n-r}} \cdot F^{r-1}(x) f(x) [1 - F(x)]^{n-r}$$

Remark : ① If $r=1$, we get p.d.f. of $X_{(1)}$

$$f_1(x) = \frac{L_n}{L_0 L_{n-1}} F^0(x) f(x) [1 - F(x)]^{n-1}$$

$$f(x_{(1)}) = f_1(x) = n f(x) (1 - F(x))^{n-1}$$

(ii) If $r=n$, we get the p.d.f. of $X_{(n)}$

$$\text{i.e. } f(x_{(n)}) = f_n(x) = \frac{L_n}{L_{n-1} L_0} F^{n-1}(x) f(x) [1 - F(x)]^0$$

$$f(x_{(n)}) = n F^{n-1}(x) f(x)$$

The Distribution of Range $W = X_{(n)} - X_{(1)}$ is given as

$$g(w) = n(n-1) \int_{-\infty}^{\infty} f(x) [F(x+w) - F(x)]^{n-2} \cdot f(x+w) ; w > 0$$

CDR of W is given as

$$G(w) = P(W \leq w) = \int_0^w g(w) dw$$

$$G(w) = n \int_{-\infty}^{\infty} f(x) [F(x+w) - F(x)]^{n-1} dx$$

Ex. $x_1, x_2, \dots, x_n \stackrel{\text{i.i.d.}}{\sim} f(x) = \frac{1}{\theta}, 0 \leq x < \theta$

Find the distⁿ of $x_{(n)}$.

We know $g(x_{(n)}) = n f(x_{(n)}) [F(x_{(n)})]^{n-1}$

$$F_X(x_{(n)}) = P(X \leq x_{(n)}) = \int_0^{x_{(n)}} \frac{1}{\theta} dx = \frac{x_{(n)}}{\theta}$$

$$\therefore g(x_{(n)}) = n \cdot \frac{1}{\theta} \cdot \left[\frac{x_{(n)}}{\theta} \right]^{n-1} = \frac{n}{\theta^n} \cdot x_{(n)}^{n-1}$$

$$\boxed{g(x_{(n)}) = \frac{n}{\theta^n} x_{(n)}^{n-1}}$$

Ex. $x_1, x_2, \dots, x_n \stackrel{\text{i.i.d.}}{\sim} f(x) = e^{-(x-\theta)}; \theta < x < \infty, -\infty < \theta < \infty$

Find the distⁿ of $x_{(n)}$.

We know $g(x_{(n)}) = n f(x_{(n)}) [1 - F(x_{(n)})]^{n-1}$

$$F(x_{(n)}) = P(X \leq x_{(n)}) = \int_{\theta}^{x_{(n)}} e^{-(x-\theta)} dx = \left[\frac{e^{-(x-\theta)}}{-1} \right]_{\theta}^{x_{(n)}} \\ = 1 - e^{-(x_{(n)}-\theta)}$$

$$g(x_{(n)}) = n e^{-(x_{(n)}-\theta)} [e^{-(x_{(n)}-\theta)}]^{n-1}$$

$$\boxed{g(x_{(n)}) = n e^{-n(x_{(n)}-\theta)}}$$